

Q3 — Guardrails, Evaluation & Multi-Agent Plan

Local RAG over the Malawi public-health dataset • Ollama qwen3:8b • September 2026

1. Guardrail design

Trust boundary. User queries and retrieved passages are untrusted input. Only the fixed application policy is authoritative. Generated text is untrusted until citation and faithfulness validation pass. Retrieved passages are delimited as SOURCE blocks and never placed in the system message.

Defense in depth. Screen every chunk with deterministic high-signal patterns; optionally use a local classifier for ambiguous text. Quarantine flagged chunks, record only IDs/hashes and reasons, then verify remaining evidence. If it is insufficient, retrieve with a refined query. Corpus text cannot trigger tools or change policy.

Control	Deterministic check	Failure action
Injection screening	Pattern/classifier scan over every chunk; preserve provenance.	Quarantine, record reason, and re-verify safe evidence.
Evidence sufficiency	Verifier identifies supported facts and missing evidence.	Refine query; after two retries, return "cannot find in sources".
Mandatory citations	Every factual claim carries an exact allow-listed chunk ID.	Reject absent, malformed, or unknown citations.
No guessing	Answerer uses only verifier-approved chunks; low temperature.	Abstain when evidence cannot support the answer.
Public-health tone	General information; no personal diagnosis, prescription, or treatment plan.	Use a safe general response or abstention.

Answer contract

The Answerer states the answer briefly, attaches [chunk_id] to each factual claim, and uses only IDs supplied in the current context. It distinguishes general public-health guidance from personal medical advice. Urgent symptom questions receive a brief instruction to contact local emergency or clinical services; the system does not diagnose. If sources are absent, unsafe, conflicting, or insufficient, the exact response is "cannot find in sources".

2. Evaluation plan

Golden set. Build 18 independently reviewed questions: 8 single-source factual, 4 multi-chunk or comparison, 3 unanswerable/out-of-domain, and 3 adversarial prompt-injection cases. Store expected documents and paragraph/chunk sets, supported facts, answerable flag, multi-citation requirement, and safety label. Use 12 while tuning and keep 6 held out; version the set with the corpus snapshot.

Measure	Definition	Initial target
Recall@4	Questions where a gold chunk appears in the top four; also strict all-evidence recall.	≥ 0.85
MRR	Mean reciprocal rank of the first relevant chunk; missing evidence scores zero.	≥ 0.75
Faithfulness	Supported factual claims $\div$ all factual claims, checked against cited text.	≥ 0.90
Citation quality	ID validity, claim coverage, and whether each citation supports its claim.	100% valid; ≥ 0.95 coverage
Abstention	Precision, recall, and F1 on unanswerable questions.	$F1 \geq 0.90$
Injection safety	Block rate on attacks and false-positive rate on clean chunks.	≥ 0.95 block; ≤ 0.05 FP

Measurement and review

Run retrieval first and compute Recall@k and MRR from annotated IDs. Generate answers, parse citations, and reject unknown IDs deterministically. Split each answer into atomic claims and ask qwen3:8b, at temperature 0 with JSON output, whether each claim is directly supported. A second human reviewer audits disagreements, all safety failures, and a random sample. Report results by question type as well as overall.

These thresholds are starting release gates and should be revised after the first labeled baseline. Every corpus, prompt, model, or retrieval change reruns the held-out set. A release fails on any unsupported personal medical recommendation even when aggregate metrics pass.

3. Multi-agent workflow

Agent	Responsibility	Structured output
Retriever	Hybrid BM25 + MiniLM search with filters; accept refined query.	ranked_chunks with IDs, text, metadata and scores; query_used
Safety	Inspect query/chunks for injection or unsafe personal-medical requests.	safe_chunks, quarantined_ids, reasons, risk_flags
Verifier	Check coverage, identify gaps, and create a focused refinement.	verdict, approved_ids, missing_facts, refined_query
Answerer	Generate from approved evidence; enforce tone and exact citations.	answer, cited_ids, abstained, reason

Routing and stopping

Query → Retriever → Safety → Verifier. If sufficient, the Answerer generates once and the system validates citation IDs and claim support. If unsafe chunks are removed, the Verifier checks the reduced set. If evidence is insufficient, its refined query returns to retrieval. Permit at most two retrieval retries. Empty safe evidence, exhausted retries, invalid citations, or faithfulness below 0.90 routes to the exact abstention response. Structured state makes decisions reproducible and testable.

MLflow observability

Use one parent MLflow run per evaluation batch and a nested run per question. Log parameters: corpus/index version, Git commit, embedding and Ollama model, chunking policy, top_k, thresholds, prompt versions, and retry limit. Log metrics: Recall@k, MRR, faithfulness, citation validity/coverage, abstention, injection block/false-positive rates, retries, and latency. Save the golden-set version, sanitized agent trace, retrieved IDs/scores, verifier JSON, prompts, and failure table. Redact sensitive queries/passages and log source IDs or hashes.

Minimal implementation

The companion snippets_q3.py provides detect_prompt_injection(text), a local-Ollama faithfulness_check(answer, cited_chunks), and dependency-injected orchestration. Production integration should add schema validation, retries for transient Ollama errors, authenticated MLflow access, and reviewer-approved emergency wording.